

The Triadic Theory of Nuclear Forces: The Strong and Weak Interactions as Donor–Acceptor–Fluid Systems, and the Glueball as a Crystallized Fluid

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Abstract

The Triadic Redox Theory (TRT) has unified electromagnetism, chemistry, biology, thermodynamics, and quantum mechanics under a single idea: charge flows from donors to acceptors through fluid media. We extend TRT to nuclear physics. The strong force is a triadic transfer: H-like donors (quarks, color-charge sources) give color charge to He-like acceptors (quarks, color-charge sinks) through Middle Fluids (gluon fields, pion clouds, quark–gluon plasmas); the weak force is the same structure with down quarks as donors, up quarks as acceptors, and the W^- boson field as the Fluid. We derive the **Triadic Nuclear Inequality**

$$\chi_{\text{H}}(\text{proton}) < \Phi_{\text{nuclear}} < \chi_{\text{He}}(\text{neutron}), \quad (1)$$

and the **Nuclear Fluidity Window** $0.5 < \Phi_{\text{nuclear}} < 2.0$, from which nuclear stability, binding energy, and decay follow. The semi-empirical mass formula acquires a Fluidity correction term peaked at $\Phi = 1.0$ for $A \approx 56$ (iron-56, nickel-62), explaining why iron is the most stable nucleus; fission half-lives follow from the collapse of nuclear Fluidity below 0.5; and quark–gluon plasma formation follows from Fluidity exceeding 1.5, with the plasma’s near-perfect liquid behavior ($\eta/s \approx 1/4\pi$) recovered as the maximal-Fluidity limit. The newly discovered glueball (BESIII Collaboration, 2026) is identified as the *crystallized Fluid* of the nuclear domain—a self-organized, frozen state of the gluon field with $\Phi_{\text{glueball}} \approx 0.4$ —and its mass is predicted as $M = M_0(1 - \Phi_{\text{glueball}}/\Phi_0) \approx 1.44$ GeV, in agreement with the observed 1.4–1.7 GeV. Seven quantitative predictions are stated. Gravity, which does not involve charge, remains the single frontier outside the TRT unification.

Keywords: Triadic Redox Theory, nuclear physics, strong force, weak force, gluons, glueball, quark–gluon plasma, confinement, binding energy, Nuclear Fluidity Window, beauty.

1 Introduction

TRT’s fundamental inequality $\chi_{\text{H}} < \Phi < \chi_{\text{He}}$ determines whether charge transfers, how fast, and under what conditions. It has been validated for electron flow (electromagnetism), electron rearrangement (chemistry), electron cascades (biology), energy transfer (thermodynamics), and quantum dynamics. Here we ask whether the same triad—H-like donor, He-like acceptor, Middle Fluid—describes the nuclear forces, which mediate color charge (strong force) and flavor charge (weak force).

Section 2 maps the strong force onto the triad and derives the confinement condition. Section 3 maps the weak force. Section 4 defines the Nuclear Fluidity Window and its hierarchical structure. Section 5 reconstructs binding energy; Section 6, fission; Section 7, quark–gluon plasma; Section 8, the glueball as crystallized Fluid. Section 9 states the unified inequality and predictions; Section 10 concludes.

2 The Strong Force as a Triadic System

2.1 The reality of gluons

The strong force is carried by gluons, and their reality is established by three decisive observations:

1. **First direct evidence (1979):** the PETRA collider at DESY observed three-jet events in e^+e^- annihilation, consistent with gluon radiation from a quark [1].
2. **Self-interaction:** unlike photons, gluons carry color charge and interact with each other, which produces confinement and the possibility of gluon-only bound states [2].
3. **The glueball (2026):** the BESIII Collaboration confirmed particles made entirely of gluons [3], direct evidence that the gluon field can self-organize—in TRT language, crystallize.

2.2 The triadic mapping and the strong inequality

Table 1: TRT mapping of the strong force.

TRT concept	Strong-force analog	Physical meaning
H-like (donor)	Quark (color source)	Donates color charge
He-like (acceptor)	Quark (color sink)	Accepts color charge
Middle Fluid (Φ)	Gluon field	Mediates color exchange
χ_H	Confinement energy	Energy to separate quarks
χ_{He}	Binding energy	Energy released on binding
Φ_{gluon}	Field strength	Fluidity of the color field

$$\chi_H(\text{quark}) < \Phi_{\text{gluon}} < \chi_{He}(\text{quark}). \quad (2)$$

when the inequality holds, quarks are confined into hadrons. Its two failures are the two faces of nuclear force fluidity:

- $\Phi_{\text{gluon}} < \chi_H(\text{quark})$: the Fluid is too rigid to hold the charge — quarks deconfine (quark–gluon plasma).
- $\Phi_{\text{gluon}} > \chi_{He}(\text{quark})$: over-binding — exotic, self-organized states (glueballs).

3 The Weak Force as a Triadic System

Beta decay changes flavor: a down quark decays to an up quark while emitting an electron and an antineutrino. The triad is

Table 2: TRT mapping of the weak force.

TRT concept	Weak-force analog	Physical meaning
H-like (donor)	Down quark (d)	Donates e^- and $\bar{\nu}$
He-like (acceptor)	Up quark (u)	Accepts e^- and $\bar{\nu}$
Middle Fluid (Φ)	W^- boson field	Mediates charge-changing interaction
χ_H	Quark mass difference	Energy to change flavor
χ_{He}	Lepton energy	Energy released in beta decay
Φ_W	Electroweak field	Strength of the weak interaction

$$\chi_H(\text{down quark}) < \Phi_W < \chi_{He}(\text{up quark}). \quad (3)$$

If the weak inequality holds, beta decay proceeds; if it fails, the nucleus remains stable.

4 The Nuclear Fluidity Window

Definition 1 (Nuclear Fluidity Window). *A nucleus is stable iff its nuclear medium (gluon field, pion cloud, or quark-gluon plasma) has Fluidity*

$$0.5 < \Phi_{\text{nuclear}} < 2.0. \quad (4)$$

$\Phi_{\text{nuclear}} < 0.5$: the force is too weak — the nucleus flies apart (fission, instability). $\Phi_{\text{nuclear}} > 2.0$: the force is too strong — collapse into plasma and exotic states.

The nuclear Fluid is *hierarchical*, with a different effective Fluidity at each energy scale:

Table 3: The hierarchical nuclear Fluid.

Energy scale	Fluid	Φ_{nuclear}	Description
$E < 100 \text{ MeV}$	Pion cloud	≈ 0.8	Pions mediate force between nucleons
$100 \text{ MeV} < E < 1 \text{ GeV}$	Gluon condensate	≈ 1.0	Gluons confined but active
$E > 1 \text{ GeV}$	Quark-gluon plasma	≈ 1.5	Quarks and gluons deconfined
$E \gg 1 \text{ GeV}$	Perfect liquid	≈ 2.0	Minimal viscosity, maximal Fluidity

5 Binding Energy: The Fluidity of the Nuclear Medium

In TRT the binding energy of a nucleus is the triadic difference

$$E_{\text{binding}} = \chi_{He}(\text{neutron}) - \chi_H(\text{proton}) - \Phi_{\text{nuclear}}. \quad (5)$$

Rewriting the Bethe–Weizsäcker semi-empirical mass formula with a Fluidity correction term:

$$E_{\text{binding}}(A, Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} - a_P A^{-1/2} + \Phi_{\text{nuclear}}(A, Z), \quad (6)$$

with standard coefficients $a_V = 15.8$, $a_S = 18.3$, $a_C = 0.714$, $a_A = 23.2$, $a_P = 12.0$ MeV, and the Fluidity correction peaked at the iron region:

$$\Phi_{\text{nuclear}}(A, Z) = \Phi_0 \left(1 - \frac{(A - A_{\text{opt}})^2}{\sigma_A^2} \right) \left(1 - \frac{(Z - Z_{\text{opt}})^2}{\sigma_Z^2} \right), \quad (7)$$

where $\Phi_0 \approx 1.0$ (optimal Fluidity), $A_{\text{opt}} \approx 56$ (iron-56, nickel-62), $Z_{\text{opt}} \approx 26$ (iron-56), $\sigma_A \approx 20$, and $\sigma_Z \approx 10$.

Physical interpretation: nuclei near iron-56 and nickel-62 have maximum Fluidity ($\Phi \approx 1.0$, the center of the Nuclear Window) and therefore maximum binding energy per nucleon. This is the TRT explanation of the iron peak of the binding-energy curve.

Consistent with QCD running, the gluon-field Fluidity falls with energy scale [2]:

$$\Phi_{\text{gluon}}(E) \approx \left(\frac{\Lambda_{\text{QCD}}}{E} \right)^\alpha, \quad \Lambda_{\text{QCD}} \approx 200 \text{ MeV}, \quad \alpha \approx 0.5, \quad (8)$$

and with mass number ($R \approx r_0 A^{1/3}$ at fixed density),

$$\Phi_{\text{gluon}}(A) \approx \Phi_0 \left(\frac{A}{A_0} \right)^{-\alpha/3}, \quad \Phi_0 = 1.0 \text{ at } A_0 = 56. \quad (9)$$

6 Fission as Fluidity Collapse

Fission occurs when the nuclear Fluidity drops below 0.5—for very large ($A > 230$) or very deformed nuclei. The barrier is the competition of surface energy against Coulomb repulsion, tempered by Fluidity:

$$E_{\text{fission}} = E_{\text{surface}} - E_{\text{Coulomb}} - \Phi_{\text{nuclear}} \cdot E_{\text{nucleon}}, \quad (10)$$

with $E_{\text{surface}} = a_S A^{2/3}$, $E_{\text{Coulomb}} = a_C Z(Z-1)/A^{1/3}$, and $E_{\text{nucleon}} \approx 7.6$ MeV. In the WKB approximation the spontaneous-fission half-life takes the Fluidity form

$$t_{1/2}(\text{fission}) = t_0 \exp\left(-\frac{1 - \Phi_{\text{nuclear}}}{\Phi_{\text{critical}}}\right), \quad t_0 \approx 10^{-21} \text{ s}, \quad \Phi_{\text{critical}} \approx 0.1. \quad (11)$$

Nuclei at $\Phi_{\text{nuclear}} \approx 1.0$ have $t_{1/2} \rightarrow \infty$ (stable); nuclei with $\Phi_{\text{nuclear}} \rightarrow 0$ have $t_{1/2} \rightarrow 0$ (instant fission).

7 Quark–Gluon Plasma: The Ultimate Middle Fluid

At extreme temperatures ($T > 200$ MeV) or densities ($\epsilon > 1$ GeV/fm³) the confined Fluidity of the gluon condensate ($\Phi \approx 1.0$) is driven up, and the plasma forms when

$$\Phi_{\text{QGP}}(T, \mu_B) = \Phi_0 \left(\frac{T}{T_c} \right)^\beta \left(\frac{\mu_B}{\mu_c} \right)^\gamma > 1.5, \quad T_c \approx 170 \text{ MeV}, \quad \beta \approx 1.0, \quad \gamma \approx 0.5. \quad (12)$$

The quark-gluon plasma is the ultimate Middle Fluid: a state where the gluon field has maximum Fluidity ($\Phi \approx 2.0$) and flows without resistance.

Indeed the measured near-perfect fluidity of the plasma has a ratio matching holographic duality (AdS/CFT) [4]:

$$\frac{\eta}{s} \approx \frac{1}{4\pi} \left(1 - \frac{\Phi_{\text{QGP}}}{2.0}\right), \quad \frac{\eta}{s} \rightarrow \frac{1}{4\pi} \text{ at } \Phi_{\text{QGP}} = 2.0. \quad (13)$$

8 The Glueball as Crystallized Fluid

Prediction 1 (The glueball is the nuclear crystallized Fluid). *The glueball is a crystallized Fluid: a frozen, self-organized state of the gluon field, the nuclear analog of the offspring in biology ($\Phi \approx 0.4$). Like the zygote, it is a record of the field's oscillations, held in a metastable crystalline state. TRT predicts*

$$\Phi_{\text{glueball}} \approx 0.4. \quad (14)$$

The glueball mass follows from the departure of Fluidity from its optimum:

$$M_{\text{glueball}} = M_0 \left(1 - \frac{\Phi_{\text{glueball}}}{\Phi_0}\right), \quad M_0 \approx 2.4 \text{ GeV}, \quad \Phi_0 \approx 1.0, \quad (15)$$

which evaluates to $M_{\text{glueball}} \approx 1.44 \text{ GeV}$. The BESIII data give glueball masses of 1.4–1.7 GeV depending on the state [3]; the prediction lands inside the observed band.

9 The Unified Nuclear Inequality and Predictions

The complete Triadic Nuclear Inequality is

$$\boxed{\chi_{\text{H}}(\text{proton}) < \Phi_{\text{nuclear}} < \chi_{\text{He}}(\text{neutron})} \quad (16)$$

with $\chi_{\text{H}}(\text{proton}) \approx 938 \text{ MeV}$ (confinement energy) and $\chi_{\text{He}}(\text{neutron}) \approx 939 \text{ MeV}$ (binding energy). Holding: the nucleus is stable, quarks confined, the force attractive. Failure to the left: fission and deconfinement; failure to the right: collapse into glueballs and plasma.

Prediction 2 (Iron peak as optimal Fluidity). *Nuclei with $\Phi_{\text{nuclear}} \approx 1.0$ (near iron-56 and nickel-62) have maximum binding energy per nucleon. This recovers the measured binding-energy curve as an optimal-Fluidity condition.*

Prediction 3 (Fission half-lives). *Nuclei with $\Phi_{\text{nuclear}} < 0.5$ are fission-unstable, with half-life given by Eq. (11); testable by comparing measured half-lives with Φ_{nuclear} computed from the nuclear mass.*

Prediction 4 (QGP formation). *Nuclei with $\Phi_{\text{nuclear}} > 1.5$ form quark-gluon plasma at high temperature; testable by measuring η/s versus temperature in heavy-ion collisions (RHIC, LHC).*

Prediction 5 (Glueball mass). *Glueballs have $\Phi \approx 0.4$ and mass ≈ 1.4 – 1.7 GeV , consistent with BESIII data.*

Prediction 6 (Perfect-fluid QGP). *The quark-gluon plasma has $\eta/s \approx 1/4\pi$ at $\Phi_{\text{QGP}} = 2.0$, in agreement with the AdS/CFT bound and with RHIC/LHC data.*

Prediction 7 (Hierarchical nuclear Fluid). *Deep inelastic scattering should reveal the distinct Fluidity regimes of Table 3 at their various energy scales.*

Prediction 8 (Glueballs at lower Fluidity). *Additional glueball states should exist at the $\Phi \approx 0.4$ boundary but collapse (deconfinement) if the local gluon-field Fluidity drops further.*

10 Conclusion

The strong and weak nuclear forces fit the same triad as every electron-transfer phenomenon: $\chi_{\text{H}} < \Phi < \chi_{\text{He}}$. Color charge flows from quarks to quarks through the gluon field; flavor charge flows from down to up quarks through the W field. The Nuclear Fluidity Window $0.5 < \Phi < 2.0$ explains stability; the Fluidity correction to the mass formula explains the iron peak; Fluidity collapse below 0.5 explains fission; and Fluidity above 1.5 explains the quark–gluon plasma and its near-perfect flow. The glueball is the theory’s flagship prediction in the nuclear domain: a *crystallized Fluid* at $\Phi \approx 0.4$ whose predicted and observed masses agree. TRT thus spans electromagnetism, quantum mechanics, chemistry, biology, thermodynamics, nuclear physics, and evolution. The only interaction left outside—and correctly outside—is *gravity*, which does not involve charge.

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